



DEPARTMENT OF MATHEMATICS

Sub Code:	AI 34/AD 34/CI 35/CY 35	Sub:	Discrete mathematical Structures	Test:	I
Time:	09.00 am -10.00 am.	Term:	20.11.23 to 02.03.24	Marks:	30
Date:	06.01.2024	Semester:	III	Sections:	AI/AD/CI/CY

Note: Answer any TWO full questions. Each main question carries 15 marks.

Q.No.	Questions	CO's	Blooms level	Marks
1.	(a) Construct the truth table for $\neg(p \vee (q \wedge r)) \Leftrightarrow (p \vee q) \wedge (p \rightarrow r)$.	CO1	L1	5
	(b) Prove by contrapositive "If $n=ab$ where a, b are positive integers then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ ".	CO1	L3	5
	(c) Consider a set $A = \{1, 2, 3, 4\}$ and the relation $R = \{(1, 2)(2, 2)(2, 3)(1, 4)(4, 1)(4, 4)\}$ defined on A . Write the matrix of relation M_R and draw digraph of relation. Also find in-degree and out-degree of all vertices.	CO2	L3	5
2.	(a) Define anti-symmetric relation. Let Z be the set of all integers and $R = \{(a, b) / a < b\}$. Discuss whether R is reflexive, anti-symmetric and transitive.	CO2	L2	5
	(b) Prove by Mathematical induction " $11^n - 4^n$ is divisible by 7, $\forall n \in \mathbb{Z}^+$ ".	CO1	L4	5
	(c) Write the converse, inverse and contrapositive of "If n is not a prime number then it is a composite number".	CO1	L3	5
3.	(a) Define Poset. Let $A = \{1, 2, 3, 4, 6, 12\}$ and on A define the relation R by aRb if a divides b . Verify (A, R) is a Poset or not?	CO2	L4	5
	(b) Define equivalence relation. The relation R defined on the set of all integers Z as $R = \{(x, y) / x, y \in \mathbb{Z}, \text{ and } (x - y) \text{ is an even integer}\}$. Verify that R is an equivalence relation or not?	CO2	L2	5
	(c) Without constructing the truth table prove $[\neg p \wedge (p \vee q)] \rightarrow q$ is a Tautology.	CO1	L3	5



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Sub Code:	AI 34/AD 34/CI 35/CY 35	Sub:	Discrete mathematical Structures	Test:	I
Time:	9.30 am to 10.30 am	Term:	31.10.2022 to 31.01.2023	Marks:	30
Date:	20.12.2022	Semester:	III	Sections:	AI/AD/CI/CY

Note: Answer any TWO full questions. Each main question carries 15 marks.

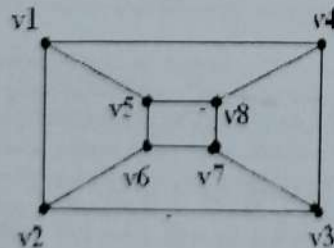
Q.No.	Questions	CO's	Blooms level	Marks
1.	(a) Show that $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent by using truth tables.	CO1	L1	5
	(b) Prove by contradiction "If $3n+2$ is odd then n is odd"	CO1	L3	5
	(c) Prove by Mathematical induction for any positive integer n , 7 divides $3^{2n+1} + 2^{n+2}$.	CO1	L3	5
2.	(a) Define equivalence relation. If R is a relation on Z , the set of all integers such that xRy if and only if $(x-y)$ is divisible by 5. Check whether R is an equivalence relation or not?	CO2	L2	5
	(b) Consider the sets $A = \{a, b, c\}$, $B = \{1, 2, 3\}$ and the relations $R = \{(a, 1), (b, 1), (c, 2), (c, 3)\}$ and $S = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$ from A to B . Determine $\bar{R}$, $\bar{S}$, R^{-1} and S^{-1} . Write their matrix representations.	CO2	L4	5
	(c) Consider a set $A = \{a, b, c, d\}$ and the relation $R = \{(a, b), (b, b), (b, d), (c, b), (c, d), (d, a), (d, c)\}$ defined on A . Write the matrix of the relation and draw the digraph of the relation.	CO2	L3	5
3.	(a) "If x is odd then $x^2 - 1$ is even" where the domain is all integers. Symbolize the statement, write the negation of the statement symbolically, write it in words.	CO1	L4	5
	(b) Define anti-symmetric relation. Let Z be the set of all integers and $R = \{(a, b) / a < b\}$. Discuss whether R is reflexive, anti-symmetric and transitive.	CO2	L2	5
	(c) Without constructing the truth table prove $p \rightarrow (q \vee r) \equiv \neg r \rightarrow (p \rightarrow q) \equiv (p \wedge \neg q) \rightarrow r$.	CO1	L3	5



DEPARTMENT OF MATHEMATICS

Sub Code:	AI34/AD34/CI/CY 35	Sub:	Discrete mathematical Structures	Test:	II
Time:	10 am to 11 am	Term:	30.11.2022 to 10.03.2023	Marks:	30
Date:	16.02.2023	Semester:	I	Sections:	AI/AD/CI/CY

Note: Answer any TWO full questions. Each main question carries 15 marks.

Q.No.	Questions	Bloom's level	CO's	Marks
1.	(a) Prove that D_{42} denotes the set of all divisors of 42 is a complemented lattice.	L2	CO2	5
	(b) Solve the recurrence relation $a_n = a_{n-1} + a_{n-2}$, given $a_1 = 2, a_2 = 7$.	L3	CO3	5
	(c) Let G be a semigroup of positive real numbers under multiplication and G' is a semigroup of all real numbers under addition. The mapping $f: G \rightarrow G'$ is defined by $f(a) = \log a, \forall a \in G$. Prove that $f: G \rightarrow G'$ is an isomorphism.	L1	CO5	5
2.	(a) A certain question paper contains three parts A, B, C with four questions in part A, five questions in part B and six questions in part C. It is required to answer seven questions selecting atleast two questions from each part. In how many different ways can a student select his seven questions for answering.	L4	CO3	5
	(b) Define semi group. Let G be the set of all non-zero real number and let $a * b = \frac{ab}{5}$. Show that $(G, *)$ is a semigroup.	L3	CO5	5
	(c) Find the graphs for which the adjacency matrices $G_1 = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$ $G_2 = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$. And check whether they are isomorphic or not.	L4	CO4	5
3.	(a) Define monoid. Verify that the set of all integers Z is a monoid under the binary operation $*$ is defined by $a * b = a + b + 1$	L2	CO5	5
	(b) (i) Define trail and path of a graph. (ii) Verify the Euler's formula for the plane graph given bellow:	L1	CO4	5
				
	(c) Find the Explicit formula for the sequence defined by $a_n = a_{n-1} + n, a_1 = 4$.	L3	CO3	5